CO3 – AT3: Resolution Algorithm Implementation

Complete Solutions – Artificial Intelligence (CSA17)

Question 1 – Rain and Wet Ground

Let R = It is raining; W = Ground is wet.

1. Propositional Logic: R → W, R. Goal: W.
2. CNF: R → W ≡ ¬R ∨ W. Clauses: C1 = ¬R ∨ W; C2 = R. Negated goal: C3 = ¬W.

3. Resolution: Resolve (¬R ∨ W) with R to obtain W. Resolve W with ¬W to obtain ■.

Conclusion: Since the empty clause ■ is derived, the goal is proved. Therefore, the ground is wet.

Question 2 – Student Assignment Submission

Let S = Rahul submitted the assignment; M = Rahul receives internal marks.

1. Propositional Logic: S → M, S. Goal: M.
2. CNF: S → M ≡ ¬S ∨ M. Clauses: C1 = ¬S ∨ M; C2 = S. Negated goal: C3 = ¬M.

3. Resolution: Resolve (¬S ∨ M) with S to obtain M. Resolve M with ¬M to obtain ■.

Conclusion: Rahul receives internal marks.

Question 3 – Library Membership

Let L = Priya is a library member; B = Priya can borrow books.

1. Propositional Logic: L → B, L. Goal: B.
2. CNF: L → B ≡ ¬L ∨ B. Clauses: C1 = ¬L ∨ B; C2 = L. Negated goal: C3 = ¬B.

3. Resolution: Resolve (¬L ∨ B) with L to obtain B. Resolve B with ¬B to obtain ■.

Conclusion: Priya can borrow books.

Question 4 – Placement Eligibility

Let A = Arun cleared the aptitude test; P = Arun is eligible for placement.

1. Propositional Logic: A → P, A. Goal: P.
2. CNF: A → P ≡ ¬A ∨ P. Clauses: C1 = ¬A ∨ P; C2 = A. Negated goal: C3 = ¬P.

3. Resolution: Resolve (¬A ∨ P) with A to obtain P. Resolve P with ¬P to obtain ■.

4. Resolution tree:

```
¬A ∨ P + A → P
P + ¬P → ■
```

Conclusion: Arun is eligible for placement.

Question 5 – Access Control System

Let P = Correct password entered; A = User is authenticated; G = User is granted access.

1. Propositional Logic: P → A, A → G, P. Goal: G.

2. CNF: $P \rightarrow A \equiv \neg P \vee A$; $A \rightarrow G \equiv \neg A \vee G$. Clauses: $C1 = \neg P \vee A$; $C2 = \neg A \vee G$; $C3 = P$. Negated goal: $C4 = \neg G$.

3. Resolution Step 1: $(\neg P \vee A) + P \rightarrow A$.

Step 2: $A + (\neg A \vee G) \rightarrow G$.

Step 3: $G + \neg G \rightarrow \blacksquare$.

4. Resolution tree: $(\neg P \vee A) + P \rightarrow A$; $A + (\neg A \vee G) \rightarrow G$; $G + \neg G \rightarrow \blacksquare$.

Conclusion: The user is granted access.

Final Answers

Question	Final Result
Q1	Ground is wet
Q2	Rahul receives internal marks
Q3	Priya can borrow books
Q4	Arun is eligible for placement
Q5	User is granted access